

Characteristic Classes

Book Notes of *Characteristic Classes*

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1 Chern Numbers & Pontrjagin Numbers

1.1 Chern Numbers

Definition 1. Let K^n be a compact complex manifold, $\dim_{\mathbb{C}} K = n$. For each partition $I = i_1, \dots, i_r$ of n , we define the I -th Chern number to be the integer

$$c_I[K^n] = c_{i_1} \cdots c_{i_r}[K^n] := \langle c_{i_1}(\tau) \cdots c_{i_r}(\tau), \mu_{K^n} \rangle,$$

where μ_{K^n} is the fundamental homology class w.r.t. the orientation determined by the complex structure.

If I is a partition of other number, we define $c_I[K^n]$ to be 0.

Example 2. For $K^n = \mathbb{C}P^n$, we have

$$\begin{aligned} c_I[\mathbb{C}P^n] &= \langle c_{i_1}(\tau) \cdots c_{i_r}(\tau), \mu_{\mathbb{C}P^n} \rangle \\ &= \left\langle \binom{n+1}{i_1} \cdots \binom{n+1}{i_r} a^n, \mu_{\mathbb{C}P^n} \right\rangle \\ &= \binom{n+1}{i_1} \cdots \binom{n+1}{i_r}. \end{aligned}$$

1.2 Pontrjagin Numbers

Definition 3. Let M^{4n} be a compact oriented smooth manifold of dimension $4n$. For each partition $I = i_1, \dots, i_r$ of n , we define the I -th Pontrjagin number to be the integer

$$p_I[M^{4n}] = p_{i_1} \cdots p_{i_r}[M^{4n}] := \langle p_{i_1}(\tau) \cdots p_{i_r}(\tau), \mu_{M^{4n}} \rangle,$$

where $\mu_{M^{4n}}$ is the fundamental homology class w.r.t. the given orientation.

If I is a partition of other number, we define $p_I[M^{4n}]$ to be 0.

Example 4. For $M^{4n} = \mathbb{C}P^{2n}$, we have

$$\begin{aligned} p_I[\mathbb{C}P^{2n}] &= \langle p_{i_1}(\tau) \cdots p_{i_r}(\tau), \mu_{\mathbb{C}P^{2n}} \rangle \\ &= \left\langle \binom{2n+1}{i_1} \cdots \binom{2n+1}{i_r} a^{2n}, \mu_{\mathbb{C}P^{2n}} \right\rangle \\ &= \binom{2n+1}{i_1} \cdots \binom{2n+1}{i_r}. \end{aligned}$$

Oriented reversing map. If we reverse the orientation of M^{4n} , since the definition of Pontrjagin classes do not depend on the orientation, we have the following

$$p_k(M) \mapsto p_k(M),$$

$$\begin{aligned}\mu_M &\mapsto -\mu_M, \\ p_I[M] &\mapsto -p_I[M].\end{aligned}$$

Thus the sign of Pontrjagin numbers change when reversing the orientation of M . If M has a nonzero Pontrjagin number, M cannot possess any orientation reversing diffeomorphism.

Manifold as a boundary. If M^{4n} is a boundary of some smooth, compact, oriented $(4n + 1)$ -dimensional manifold-with-boundary. Then all Pontrjagin numbers of M must be zero. Thus if some $p_I[M] \neq 0$, then M cannot be the boundary of such manifold.

1.3 Symmetric Functions

We know that all symmetric polynomials in $\mathbb{Z}[t_1, \dots, t_n]$ is itself a polynomial ring:

$$\mathcal{S} = \mathbb{Z}[\sigma_1, \dots, \sigma_n] \subset \mathbb{Z}[t_1, \dots, t_n],$$

where σ_i is the i -th elementary symmetric polynomial.

$$1 + \sigma_1 + \dots + \sigma_n = (1 + t_1) \cdots (1 + t_n).$$

If we focus on the symmetric polynomials of degree k , denoted by $\mathcal{S}^k$, clearly it is a free abelian group with basis

$$\sigma_{i_1} \cdots \sigma_{i_r}, \quad I = i_1, \dots, i_r \text{ is a partition of } k.$$

For some reason, it is more convenient to use another basis of $\mathcal{S}^k$, which is described as follows.

Define two monomials in $\mathbb{Z}[t_1, \dots, t_n]$ to be equivalent if they differ by a permutation of the variables. We define

$$\sum t_1^{a_1} \cdots t_r^{a_r}$$

to be the summation of all distinct monomials equivalent to $t_1^{a_1} \cdots t_n^{a_n}$. Clearly, it is a symmetric polynomial.

Lemma 5.

$$\sum t_1^{a_1} \cdots t_r^{a_r}$$

where $a_1 \cdots a_r$ ranges over all partitions of k with length $r \leq n$, form another basis of $\mathcal{S}^k$.

Now for each partition $I = i_1, \dots, i_r$ of k , we define a polynomial s_I in k variables as follows: choose $n \geq k$, define

$$s_I(\sigma_1, \dots, \sigma_k) := \sum t_1^{i_1} \cdots t_r^{i_r},$$

both sides are symmetric polynomials in n variables.

Notes that s_I does not depend on the choice of n , for if we choose another $n' > n$, then we can set $t_{n+1} = \dots = t_{n'} = 0$ to get back to the previous case.

Clearly, all s_I form a basis of $\mathcal{S}^k$ as I ranges over all partitions of k .

Now let's look at the cohomology ring $H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z})$. the splitting map

$$\phi : \mathbb{C}P^\infty \times \cdots \times \mathbb{C}P^\infty \rightarrow \mathrm{Gr}_n(\mathbb{C}^\infty)$$

induces an injective homomorphism of cohomology rings

$$\phi^* : H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z}) \rightarrow H^*(\mathbb{C}P^\infty \times \cdots \times \mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}[t_1, \dots, t_n],$$

where t_i is the generator of the i -th factor. ϕ^* maps $H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_n]$ isomorphically onto the symmetric polynomial subring $\mathcal{S} \subset \mathbb{Z}[t_1, \dots, t_n]$, with c_i mapped to σ_i .

Since $\phi^* s_I(c_1, \dots, c_k) = s_I(\sigma_1, \dots, \sigma_k)$ forms a basis of $\mathcal{S}^k$ as I ranges over all partitions of k , by the injectivity of ϕ^* , $s_I(c_1, \dots, c_k)$ also forms a basis of $H^{2k}(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z})$.

1.4 A Product Formula

Let ω be an n -dimensional complex vector bundle over a compact complex manifold B . For each $k \geq 0$ and partition $I = i_1, \dots, i_r$ of k , we define

$$s_I(c(\omega)) := s_I(c_1(\omega), \dots, c_k(\omega)) \in H^{2k}(B; \mathbb{Z}).$$

sometimes we simply write it as $s_I(c)$.

Lemma 6 (Thom).

$$s_I(c(\omega \oplus \omega')) = \sum_{J, K} s_J(c(\omega)) s_K(c(\omega')),$$

where the summation is taken over all partitions J, K with juxtaposition JK equal to I .

Remark. We can view this formula as a generalization of the Whitney sum formula.

Proof. Let $\phi : F(B) \rightarrow B$ be the flag manifold of $\omega \oplus \omega'$, with the splitting bundle

$$\phi^*(\omega) = \eta_1 \oplus \cdots \oplus \eta_n, \quad \phi^*(\omega') = \eta'_1 \oplus \cdots \oplus \eta'_{n'}.$$

Then we have

$$\begin{aligned} \phi^*c(\omega) &= (1 + c_1(\eta_1)) \cdots (1 + c_1(\eta_n)) =: (1 + t_1) \cdots (1 + t_n), \\ \phi^*c(\omega') &= (1 + c_1(\eta'_1)) \cdots (1 + c_1(\eta'_{n'})) =: (1 + t_{n+1}) \cdots (1 + t_{n+n'}). \end{aligned}$$

Thus

$$\begin{aligned} \phi^*s_I(c(\omega \oplus \omega')) &= \sum t_1^{i_1} \cdots t_r^{i_r} \\ &= \sum_{\substack{JK=I, \\ J=j_1, \dots, j_{r_1}, \\ K=k_1, \dots, k_{r_2}}} \left(\sum t_1^{j_1} \cdots t_{r_1}^{j_{r_1}} \right) \left(\sum t_{n+1}^{k_1} \cdots t_{n+r_2}^{k_{r_2}} \right) \\ &= \sum_{JK=I} s_J(c(\eta_1 \oplus \cdots \oplus \eta_n)) s_K(c(\eta'_1 \oplus \cdots \oplus \eta'_{n'})) \\ &= \sum_{JK=I} s_J(\phi^*c(\omega)) s_K(\phi^*c(\omega')) \\ &= \phi^* \left(\sum_{JK=I} s_J(c(\omega)) s_K(c(\omega')) \right). \end{aligned}$$

Since ϕ^* is injective, we get the desired formula. $\square$

Now for a compact complex manifold K^n . For each partition $I = i_1, \dots, i_r$ of n , we denote

$$s_I[K^n] = s_I(c)[K^n] := \langle s_I(c(\tau)), \mu_{K^n} \rangle \in \mathbb{Z}.$$

This characteristic number is clearly a linear combination of Chern numbers.

Corollary 7.

$$s_I[K^m \times L^n] = \sum_{JK=I} s_J[K^m] s_K[L^n].$$

where the summation is taken over all partitions J of m and K of n with juxtaposition JK equal to I .

Proof. Since $\tau_{K \times L} = \tau_K \times \tau_L = \pi_1^* \tau_K \oplus \pi_2^* \tau_L$, where π_1, π_2 are the projections from $K \times L$ to K and L respectively,

$$s_I[K \times L] = \langle s_I(c(\tau_{K \times L})), \mu_{K \times L} \rangle$$

$$\begin{aligned}
&= \sum_{JK=I} \langle s_J(c(\pi_1^* \tau_K)) s_K(c(\pi_2^* \tau_L)), \mu_K \times \mu_L \rangle \\
&= \sum_{JK=I} \langle s_J(c(\tau_K)), \mu_K \rangle \langle s_K(c(\tau_L)), \mu_L \rangle \\
&= \sum_{JK=I} s_J[K] s_K[L].
\end{aligned}$$

□

Corollary 8. *As a special case, $s_{m+n}[K^m \times L^n] = 0$, for $m, n > 0$.*

There are analogous results for Pontrjagin classes and numbers. Let ξ be a real vector bundle over a compact smooth manifold B . For each $k \geq 0$ and partition $I = i_1, \dots, i_r$ of k , we define

$$s_I(p(\xi)) := s_I(p_1(\xi), \dots, p_k(\xi)) \in H^{4k}(B; \mathbb{Z}).$$

There are similar product formulas for Pontrjagin classes

$$s_I(p(\xi \oplus \xi')) \equiv \sum_{J,K} s_J(p(\xi)) s_K(p(\xi')),$$

modulo elements of 2-torsions. Since pairing with the fundamental class annihilates all 2-torsions, we have

$$s_I[M \times N] = \sum_{JK=I} s_J[M] s_K[N],$$

Notes that these characteristic numbers will be zero unless both the dimensions of M and N are divisible by 4.

1.5 Linear Independence of Chern Numbers and of Pontrjagin Numbers

Theorem 9 (Thom). *Let $K^1, \dots, K^n$ be compact complex manifolds with $s_k(c)[K^k] \neq 0$. Then the $p(n) \times p(n)$ matrix*

$$\left(c_{i_1} \cdots c_{i_r} [K^{j_1} \times \cdots \times K^{j_s}] \right)$$

is nonsingular, where both $I = i_1, \dots, i_r$ and $J = j_1, \dots, j_s$ range over all partitions of n .

Similarly, we have the following theorem for Pontrjagin numbers.

Theorem 10 (Thom). *Let $M^4, \dots, M^{4n}$ be compact oriented smooth manifolds with $s_k(p)[M^{4k}] \neq 0$. Then the $p(n) \times p(n)$ matrix*

$$\left(p_{i_1} \cdots p_{i_r} [M^{4j_1} \times \cdots \times M^{4j_s}] \right)$$

is nonsingular, where both $I = i_1, \dots, i_r$ and $J = j_1, \dots, j_s$ range over all partitions of n .

Proof. We only prove the first theorem, the second one is completely analogous.

Since the basis $c_{i_1} \cdots c_{i_r}$ isn't well behaved under the product operation, we change to the basis $s_I(c)$. By the product formula,

$$s_I[K^{j_1} \times \cdots \times K^{j_s}] \neq 0$$

only if $I = i_1, \dots, i_r$ is a refinement of $J = j_1, \dots, j_s$. Thus by suitable ordering of both the rows and columns, we can assume that the matrix is upper triangular. The diagonal elements are

$$s_{i_1, \dots, i_r} [K^{i_1} \times \cdots \times K^{i_r}] = s_{i_1} [K^{i_1}] \cdots s_{i_r} [K^{i_r}] \neq 0.$$

Thus the matrix is nonsingular. □